James Machmouchi

Software Engineer · Full-stack web & mobile

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PROFILE

Software engineer who builds and ships full-stack products on my own, from the database to the screen. I've built a live meal-planning SaaS with Stripe subscriptions, two iPhone apps, a cross-platform React Native app and a Qur'an companion that answers how you feel. I'm also an HCPC-registered NHS dietitian. At work I've built tools that estimate patients' nutritional intake from food charts and diaries (one saves around 10–15 minutes per outpatient assessment), and I'm the team's Copilot champion. I bring real clinical and nutrition knowledge to health products, plus years of explaining complex things clearly to patients, doctors and teams.

PROJECTS

BiteSync · bitesync.io · live 2026 – now

Meal-planning web app that builds recipes, plans the week and writes the grocery list.

- Designed and built the whole product: macro-aware planning against a personal TDEE target, meal templates for each goal, and grocery lists
- Stripe subscriptions and webhooks, Supabase auth with row-level security, and a blog with SEO comparison posts
- Shipped through 100+ merged pull requests with CI on GitHub

`Next.js · React · TypeScript · Supabase · Stripe · Vercel`

Dua & Ayah Companion · web app 2026 – now

Users write how they feel and get a matching Qur'anic ayah or dua, with translation and recitation.

- Wrote a tested intent matcher that weighs phrases over single words and handles negation, so "I'm not sad" doesn't match sadness
- Qur'an reader, prayer times and daily adhkar; installable PWA with a content-review workflow and a premium tier enforced in the database

`Next.js · TypeScript · Supabase · Tailwind · Vitest`

Greetings From · iPhone & Android 2026

Camera app that turns a photo into a vintage postcard: the place name in 3D letters with the photo showing through, and a back that fills in the postmark and a landmark stamp.

- Custom drawing with Skia, exported at print resolution; 3D card flip with Reanimated
- No backend: photos never leave the phone

`React Native · Expo · TypeScript · Skia · Reanimated`

Nafas · iPhone app 2026 – now

Arabic-first app to help adults quit vapes, cigarettes and shisha, with a Ramadan mode.

- Full right-to-left layout, StoreKit 2 subscriptions, and Home and Lock Screen widgets
- Business logic in plain Swift with unit tests; all data stays on the device

`SwiftUI · StoreKit 2 · WidgetKit · XcodeGen`

Practise to Unlock · iPhone & iPad prototype 2026

Keeps chosen apps locked until the microphone hears real instrument practice, using Apple's Screen Time APIs and on-device pitch detection.

`SwiftUI · Family Controls · AVAudioEngine`

AgentZero · team project 2026

WhatsApp AI copilot for Dubai estate agents. My part:

- Built instant callbacks: an AI voice agent calls website leads within 60 seconds
- Moved the assistant onto Meta's Muse model with Claude as a fallback, and wrote cost tests comparing the two

`Next.js · Supabase · Twilio · WhatsApp Cloud API · Vapi`

SKILLS

Languages
TypeScript, JavaScript, Swift, SQL, HTML, CSS, Python

Frontend & mobile
Next.js, React, React Native, Expo, SwiftUI, Tailwind

Backend & data
Supabase, Postgres, row-level security, Redis, Convex

Payments & APIs
Stripe, StoreKit 2, Twilio, WhatsApp Cloud API, Vapi, Resend

AI
Claude and Meta Muse APIs, AI voice agents

Tools
Git, GitHub Actions, Vercel, Vitest, Xcode, Figma

EDUCATION

MSc Dietetics, Merit
Birmingham City University · 2022 – 2024

BSc Nutrition, 2:1
University of Hertfordshire · 2017 – 2020

ALSO

HCPC-registered Dietitian

Languages
English and Arabic, both fluent

WORK EXPERIENCE

Dietitian · Lewisham University Hospital, NHS — Aug 2024 – now

- Built a tool that estimates calories and macros from inpatients' food charts and checks nutritional adequacy, so oral nutritional supplements (ONS) are prescribed correctly
- Built a second tool that estimates outpatients' intake from food diaries, saving around 10–15 minutes per assessment
- Copilot champion for the team, helping colleagues use AI in their day-to-day work
- Run outpatient clinics and inpatient care for complex cases, seeing 50+ patients a week
- Ran audits that found £2,000+ a year in potential savings
- Teach junior doctors and staff, and supervise dietetic students

Student Dietitian · Birmingham Community Healthcare — 2022 – 2023

Clinical placements in a team of 20; ran service audits and delivered presentations to patients and staff.

Nutritionist · Nutrition Enhancement Academy — 2020

One-to-one nutrition advice and goal setting for clients, in person and online.

Assistant Manager · The Banc Restaurant — 2013 – 2017

Managed a team of 15, handled stock and complaints, and helped grow revenue against owner targets.